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MILITARY SPECIFICATION

POWER SUPPLY, HYDRAULIC AND ELECTRIC, PORTABLE

This specification is approved for use within the U.S. Army Armament, Munitions and Chemical Command, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers one type of portable (wheeled) hydraulic and electric power supply unit for ground servicing of aircraft weapon systems. The complete Power Supply consists of three major components referred to herein as follows:

Power Supply	-	Drawing F12007000
Distribution Box	-	Drawing F12007200
Battery	-	MS24498

2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications, standards and handbooks. The following specifications, standards and handbooks form a part of this specification to the extent specified herein. Unless otherwise specified, the issues of these documents shall be those listed in the issue of the Department of Defense Index of Specifications and Standards (DODISS) and supplement thereto, cited in the solicitation (see 6.2).

SPECIFICATIONS

FEDERAL

BB-N-411	Nitrogen, Technical

MILITARY

MIL-C-104	Crate, Wood, Lumber and Plywood Sheathed, Nailed and Bolted

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Commander, U.S. Army ARDEC, ATTN: SMCAR-BAC-S, Picatinny Arsenal, New Jersey 07806-5000 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

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MIL-P-116	Preservation, Methods of
MIL-T-152	Treatment, Moisture and Fungus Resistant, of Communications, Electronic and Associated Electrical Equipment
MIL-P-6063	Packaging of Batteries, Storage, Charged and Dry - Uncharged and Moist, General Specification for
MIL-F-13926	Fire Control Material Manufacture and Inspection General Specification for
MIL-P-14232	Parts, Equipment and Tools for Army Material, Packaging and Packing
MIL-H-50606	Hydraulic Fluid, Petroleum Base, Aircraft, Missile and ordnance
MS-24498	Battery, Aircraft Storage, Nickel-Cadmium, 24 volts, 34 Ampere Hour, 27 Degree C

STANDARDS

MILITARY

MIL-STD-109	Quality Assurance Terms and Definitions.
MIL-STD-129	Marking for Shipment and Storage
MIL-STD-454	Standard General Requirements for Electronic Equipment
MIL-STD-461	Electromagnetic Emission and Susceptibility Requirements for the Control of Electromagnetic Interference
MIL-STD-810	Environmental Test Methods
MIL-STD-1186	Cushioning, Anchoring, Bracing, Blocking, and Waterproofing with Appropriate Test Methods
MIL-STD-45662	Calibration System Requirements
MIL-STD-2000	Standard Requirements for Soldering Electrical and Electronic Assemblies

(Unless otherwise indicated, copies of federal and military specifications, standards, and handbooks are available from the Standardization, Documents Order Desk, Bldg. 4D, 700 Robbins Avenue, Philadelphia, PA 19111-5094.)

2.1.2 Other Government documents, drawings, and publications. The following other Government documents, drawings, and publications form a part of this specification to the extent specified herein. Unless otherwise specified, the issues shall be those in effect on the date of the solicitation (see 6.2).

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DRAWINGS (see 6.4)

US ARMY ARMAMENT RESEARCH, DEVELOPMENT AND ENGINEERING CENTER (ARDEC)

A5910821	Power Supply, Hydraulic and Electric, Portable
F12007000	Power Supply, Hydraulic and Electric, Portable
F12007200	Distribution Box Assembly

PUBLICATIONS

Dept. of Army Technical Manual

TM9-4933-211-14	Power Supply, Hydraulic and Electric, Portable

(Copies of other government documents, drawings, and publications required by the suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Non-Government publications. The following document(s) form a part of this specification to the extent specified herein. Unless otherwise specified, the issues of the documents which are DOD adopted shall be those listed in the issue of the DODISS specified in the solicitation. Unless otherwise specified, the issues of documents not listed in the DODISS are the issues of the documents cited in the solicitation (see 6.2).

AMERICAN SOCIETY FOR TESTING AND MATERIALS (ASTM)

ASTM-D-3951 - Commercial Packaging, Standard Practice For

(Application for copier should be addressed to the American Society for Testing and Materials, 1916 Race Street, Philadelphia, PA 19103-1187.)

National Fire Protection Association Pamphlet 70 - National Electrical Code. (Application for copies should be addressed to the National Fire Protection Association, One Batterymarch Park, P.O. Box 9101, Quincy, MA 12269-9101.)

2.3 Order of precedence. In the event of conflict between the text of this specification and the references cited herein (except for related associated detail specifications, specification sheets or MS standards), the text of this specification shall take precedence. Nothing in this specification, however, shall supersede applicable laws and regulations unless a specific exemption has been obtained.

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3. REQUIREMENTS

3.1 Specification sheets. The individual item requirements shall be specified herein and in accordance with the applicable MS sheets. In the event of any conflict between requirements of this specification and the MS sheets, the latter shall govern.

3.2 First article. Requirements for submission of First Article shall be as specified in the contract (see 6.2). Unless otherwise specified (see 6.2), the First Article shall include the pilot pack (see 5.1).

3.3 Material, construction and design. The Power Supply, Hydraulic and Electric, Portable, referred to herein as the power supply, shall conform to the material, construction and design requirements specified herein and on Drawing A5910821 and drawings pertaining thereto.

3.4 Performance. The power supply shall provide, as a minimum, a hydraulic fluid flow of 0 to 3.5 gallons per minute as a pressure of 0 to 1500 pounds per square inch gage. The DC power supply shall provide, as a minimum, 26 to 30 volts variable, regulated to ±1 percent at the power supply at a current load of 0 to 40 amperes. The internal battery charging circuit shall provide a charging current of 7-13 ampere monitored on a control panel mounted meter. External battery charging shall be available but shall not be current limited. There shall be no degradation of performance while operating under the following conditions:

a. Operating temperatures ranging from 0 deg. to +125 deg.F.
b. Operation at pressure altitudes from sea level to 6,000 feet at a temperature range of 0 deg. to 125 deg. F.
c. A relative humidity of 95 to 100 percent at an operating temperature of 125 deg. F with the power supply and distribution box control panels exposed and connected for operation as specified in 3.17.
d. Exposure to salt sea atmosphere as specified In MIL-STD-810, Method 509, with the power supply and distribution box control panels exposed and the unit connected for operation as specified in 3.17.

3.4.1 Storage temperatures. The power supply shall be capable of withstanding storage temperatures ranging from -65 deg. to +160 deg. F.

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3.5 Hydraulic system. Surfaces transmitting the hydraulic fluid shall be free of foreign material or contaminant exceeding the following limits of particle contamination. Particle count shall include only particles 5 microns in size or larger.

Particle Size Range (Microns)	Maximum Particle Per Hundred ML of Flushing Solution
0 - 15	16,000
15 - 25	2,850
25 - 50	506
50 - 100	90
over 100	16

Conformance to the contamination limits specified shall be determined by inspection samples taken from the hydraulic flushing fluid at the outlet side of the power supply.

3.5.1 High-pressure pump. An axial piston type, high-pressure pump shall be provided of sufficient rating to meet all requirements specified herein and on the applicable drawings.

3.5.2 Filters. Filters (low-pressure and high-pressure) shall be provided and located in the system as shown on the applicable drawing. The high-pressure filter shall be protected with a differential pressure switch and shall indicate when the pressure drop across the filter reaches a value of 35 $\pm$ 5 psi, at which time change of the filter element shall be required.

3.5.2.1 High-pressure filter. A high-pressure filter rated for operation at 3.5 gallons per minute and 1500 pounds per square inch gage, with replaceable element, shall be supplied according to applicable drawings. The filter element shall be rated for not less than 5 microns nominal particle size removal.

3.5.2.2 Low-pressure filter. A low-pressure filter shall be provided according to applicable drawings utilizing a 40 mesh removable strainer.

3.5.3 High-pressure relief valve. A high–pressure relief valve shall be provided meeting the requirements of Drawing 12007080 and preset to 1750 psi gage, $\pm$ 10 percent.

3.5.4 Reservoir. A reservoir assembly meeting the requirements of the applicable drawings shall be supplied.

3.5.5 External leakage. Under all operating conditions herein, there shall be no evidence of external leakage other than slight wetting insufficient to form a drop.

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3.5.6 Oil cooler. An air-to-oil cooler shall be provided meeting the requirements of the applicable drawings.

3.5.7 Accumulator. An accumulator shall be provided that meets the requirements of the applicable drawings. For test purposes, the accumulator shall be charged with 60 cubic inches of dry nitrogen, BB-N-411, Type I, Class 2, Grade B or C, at a pressure of 750 ± 25 psi gage. Prior to shipment, the accumulator shall be depressurized and drained.

3.5.8 Flow meter. A variable area glass tube type flow meter graduated in increments of .1 GPM meeting the requirements of the applicable drawings shall be provided.

3.5.9 Thermal switch. An adjustable type thermal switch shall be provided meeting the requirements of Drawing 12007153.

3.6 Electrical system. All electrical equipment and wiring shall conform to NFPA Pamphlet 70 unless otherwise specified on the applicable drawings. The electrical system shall be connected as shown on the applicable drawings.

3.6.1 Pump motor. An electric motor meeting the requirements of the applicable drawings shall be supplied. The manufacturer's recommended full load current shall not be exceeded during operation and tests.

3.6.2 Wiring. All wiring between electrical components shall be in accordance with NFPA Pamphlet 70 and Drawing 7792555. A color code or number code shall be used on all wires and terminal lugs to simplify tracing and preworking of the circuits.

3.6.3 DC power supply. A solid state DC power supply shall be provided within the power supply housing meeting the requirements of the applicable drawing. The-supply shall be adjusted to provide a minimum of 26 - 30 volts DC output with an AC power input of 208 - 230 volts AC, 3 phase, 50 - 60 Hertz. Line and load voltage regulation shall be adjusted to ± 1 percent at the power supply from O to 40 ampere load. Peak to peak ripple at full load shall not exceed 200 MVAC.

3.6.4 Battery. One aircraft type storage battery conforming to MS 24498 shall be provided by the contractor with each complete power supply unit. The battery shall be packaged and packed for shipment as specified in Section 5.

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3.7 Distribution box. The distribution box shall be supplied as a separate unit in a weather resistant case as specified on Drawing F12007200. Accessory cable and hose assemblies shall be stored in the case cover.

3.8 Internal accessibility. All doors and adjustable louvers of the power supply shall be free from binding and shall be easy to operate with one hand. The rear housing assembly shall be easily removable by two men by disconnecting fasteners and lifting or sliding the entire assembly off the remainder of the power supply assembly.

3.9 Electromagnetic interference. The power supply shall meet the requirements of MIL-STD-461 for Class IIB equipment, Table II.

3.10 Nameplate. An engraved, or etched, metal or plastic nameplate containing the information described on the applicable drawings shall be permanently attached to the hydraulic source box adjacent to the item or function it identifies.

3.11 Electrical components. All electrical components shall be mounted on the control panel as specified on the applicable drawings.

3.12 Fungus control. Electrical components and circuit elements shall be treated for fungus resistance in accordance with Type II requirements of MIL-T-152, except that the statement of treatment shall not be required.

3.13 Wiring. All wiring in the control panel and cable assembly shall be in accordance with Drawing 7792555.

3.14 Soldering. Soldering shall be in accordance with MIL-STD-454, Requirement 5.

3.15 Air transportability.

3.15.1 Tie-downs. Tie–down connections shall be provided for securing the power supply during air transport. The connections shall withstand, without weld failure or permanent deformation, the following forces applied independently:

Fore	3 g
Aft	1-1/2 g
Lateral	1-1/2 g
Up	2 g
Down	4-1/2 g

Note: One g shall be equal to the dry cargo weight of the power supply.

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The equipment shall withstand the forces applied dynamically for 0.1 second. Scientific computation and analysis may be submitted by the contractor to verify compliance to the above requirements in lieu of actual tests.

3.15.2 Altitude. The performance of the power supply shall not be affected by the effects of reduced pressure resulting from air transport at altitudes up to 15000 feet. There shall be no damage to any component and leakage shall not exceed that specified in 3.4.5.

3.16 Identification of product. An engraved, or etched, metal or plastic product identification nameplate as specified on the applicable drawing shall be permanently attached to the front of the case assembly. The serial number of the distributor box shall be engraved or etched into the front of the distribution panel to agree with that on the product identification plate.

3.17 Technical manual. The contractor shall provide one copy of TM9-4933-211-14 (furnished as Government Furnished Property) with each complete power supply. The TM shall be stored in the Cover Assembly, P/N 12007006.

3.18 Controls and operational characteristics.

3.18.1 Preparation for operation.

a. Hydraulic system. Prior to operation, the power supply shall be cleaned and flushed in accordance with Drawing 11011289. The hydraulic reservoir shall be filled with fluid conforming-to MIL-H-50606 as indicated on the reservoir level gage. Connect the supply port and corresponding hose assembly to the pressure side of a simulated load. Connect the return port and its associated hose to the return side of the simulated load.
b. AC power supply connection. Connect the AC power supply cable to a 220/240 volt AC, 3 phase, 50/60 HZ, 7.5 KVA power source. .
c. Distribution box connection. Interconnect the distribution box to the power supply as specified below.

(1) Connect the DC power supply cable to connector J105 of the distribution box.
(2) Connect the battery cable to connector J104 of the distribution box.

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d. Accumulator. Assure that the accumulator is precharged with dry nitrogen to approximately 750 psi. (see 3.4.7)

3.18.2 Starting power supply.

a. Close the outlet shutoff valve. This shall prevent flow of hydraulic oil to simulated load.
b. Open bypass valve. This shall allow fluid within the power supply to return to the reservoir.
c. Place selector switch in OFF position. This 3 position switch shall control the DC power supply output, allowing either battery charging or DC output mode of operation.
d. When the MASTER switch is In the ON position, the Green Indicator shall light.
e. When the Motor Start Switch is depressed, the hydraulic pump motor and heat exchanger fan shall operate.

3.18.3 Hydraulic operation.

a. TO adjust the pump compensator, close the bypass valve and adjust for 1500 psi as read on the high-pressure gage located on the control panel.
b. To adjust the pump volume control, open the bypass valve and adjust for the 3.5 ± .1 GPM as read on the flow meter.
c. Close bypass valve.
d. When the outlet shutoff valve is fully open, hydraulic fluid shall be applied to the simulated load.
e. During operation, if the high pressure filter becomes partially clogged causing a differential pressure across it of 35 ± 5 psi gage, the differential pressure switch shall light the red indicator located on the control panel.
f. If the temperature of the hydraulic fluid reaches 175 deg.F or higher, the power supply shall shut down automatically thru the. action of the thermostatic switch.
g. Stopping hydraulic operation.

(1) Close outlet shutoff valve.
(2) Open bypass valve.
(3) Press motor stop switch, the pump motor shall stop.

3.18.4 DC supply system operation.

a. Preliminary adjustments.

(1) Disconnect AC power supply cable cable from power source.

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(z) Remove the rear housing assembly by loosening the fasteners and lifting or sliding the entire assembly off the remainder of the power supply cabinet.
(3) Remove the DC power supply assembly cover.
(4) On the DC power supply regulator assembly, preset resistor (R1) to 20 ohms minimum.
(5) Set variable resistor (R4) at mid position.
(6) Set variable resistor (R16) at maximum resistance.
(7) Reconnect AC power supply cable to a 220/240 volt AC, 3 phase, 50/60 HZ, 7.5 KVA power source.

b. DC power supply.

(1) Output voltage adjustment.

(a) When the MASTER switch is in the ON position, the green indicator shall light.
(b) Place bolt meter SELECTOR switch on the distribution box to the 28 volt buss position.
(c) When the SELECTOR switch on the power supply control panel Is in the DC POWER SUPPLY position, the voltmeter on distribution box shall indicate the power supply output voltage.
(d) Rotate the DC voltage adjust rheostat to the maximum clockwise position. (Index line on knob lines up with battery charger mark on control panel). The voltmeter on the distribution box shall indicate a minimum of 30 volts DC.

NOTE: If voltage indication is incorrect, readjust variable resistor (R4) and lock adjustment.

(e) Rotate DC voltage adjust rheostat fully counter-clockwise. The voltmeter on the distribution box shall indicate a maximum of 26 volts DC.

NOTE: If voltage reading is not correct, readjust variable resistor (R4). Continue this process between the high and low output voltage until the required voltage range of 26 - 30 is obtained.

(2) Voltage regulator adjustment.

(a) Prepare a 40 ampere resistive load to be connected between pins H and K on connector J102.
(b) With input voltage set to 220 volts AC, readjust the DC voltage adjust rheostat to an output of 30 volts DC.
(c) Apply the 40 ampere load to connector J102 and check the regulation at no load to full load. Regulation shall be $\pm$ 1 percent or better at the power supply, TBI-3($\mp$) and TBI-4(-), as indicated by a test voltmeter.

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NOTE: If regulation is not within ± 1 percent, adjust slide wire resistor (R1) by increasing its value in small increments above the initial 20 ohm setting.

(d) With the output voltage adjusted to 28 volts DC, the peak to peak ripple shall not exceed 200 millivolts at full load.

(3) Power supply current limit.

(a) Adjust the DC voltage adjust rheostat to obtain an output of 28 volts as indicated on the distribution box voltmeter.
(b) Connect an external 24 volt DC source or battery, external load, 0 to 150 ampere ammeter as shown in Figure 1. Starting with the resistance set to maximum, increase current to 90 amperes thru resistor (R10). Observe current limit on voltmeter (voltage will drop). At 95 - 115 amperes, output shall be approximately 0 - 1 volt DC.
(c) Apply a 65 ampere load between pins H and K on connector J102 long enough to trip the 50 ampere, DC circuit breaker. After removing the load, check the operation of the DC resetting mechanism located on the control panel. Operation of the mechanism shall again cause 28 volt DC to be indicated on the distribution box voltmeter.

(c) Battery charging circuit.

(1) Battery charging voltage adjustment.

(a) Place voltmeter SELECTOR switch on the distribution box in the battery buss position. Disconnect connector P3 from J105.
(b) Place SELECTOR switch on the power supply control panel in the BATTERY CHARGE position. The voltmeter on the-distribution box shall indicate the battery charge output voltage.
(c) Rotate DC voltage adjust rheostat to the maximum clockwise position (Index line on knob lines up with battery charger mark on control panel).
(d) Connect a 5,000 mfd capacitor across pins B and E on Connector J101 in the distribution box.
(e) With control panel SELECTOR switch in the BATTERY CHARGE position, the output indicated on the distribution box voltmeter shall be 29 ± 0.5 volts DC. Re-adjust variable resistor (R16) to obtain the necessary voltage reading and lock adjustment.

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(2) Battery charging circuit current limit.

(a) Connect a 5,000 mfd capacitor and external load with a 0 to 50 ampere ammeter between pins B and E on connector J101 on the distribution box as shown in Figure 2.
(b) Increase the load. The current limit shall occur between 7 to 13 amperes.

d. Distribution box voltage measurement.

(1) Place voltmeter SELECTOR switch on the distribution box to the 28 volt DC buss position.
(2) When the selector switch on the power supply control panel is in the DC POWER SUPPLY position, the voltmeter on the distribution box shall indicate the power supply output voltage.
(3) Rotate DC voltage adjust rheostat to adjust the output voltage of the power supply to 28 volts DC as indicated on the distribution box voltmeter.
(4) When using an external voltmeter, the following DC voltage shall be at the points specified in the following table:

Positive Meter Probe	Negative Meter Probe	Voltage (± 2 VDC) Reading	CB Position
J107-A	J107-B	28 VDC	
J106-A	J106-B	28 VDC	
J103 Top Left	J103 Bottom Left	28 VDC	
J103 Top Right	J103 Bottom Right	28 VDC	
TJBPI	TJBP4	28 VDC	CB1-1N
J102-P	J102-H	28 VDC	CB1-1N
TJBP2	TJBP5	28 VDC	CB2-1N
J102-0	J102-H	28 VDC	CB2-1N
J102-R	J102-Y	28 VDC	CB3-1N
TJBP 3	TJBP6	28 VDC	CB4-1N
J102J	J102-Y	28 VDC	CB4-1N

(5) Place SELECTOR switch on the control panel to the OFF position.
(6) When the MASTER switch Is in the OFF position, the green indicator on the control panel shall extinguish.
(7) Replace all covers and housings and clean and flush the hydraulic system in accordance with Drawing B11011289.

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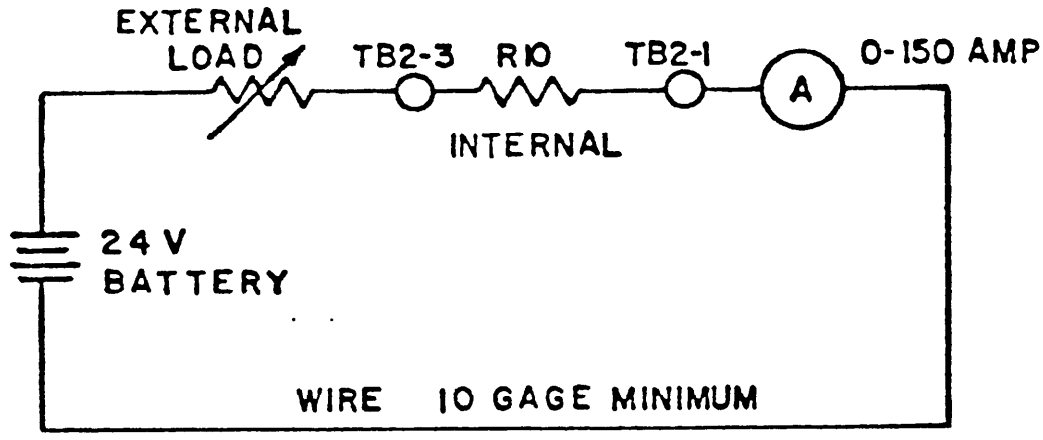


FIGURE 1

Power Supply Current Limit Circuit

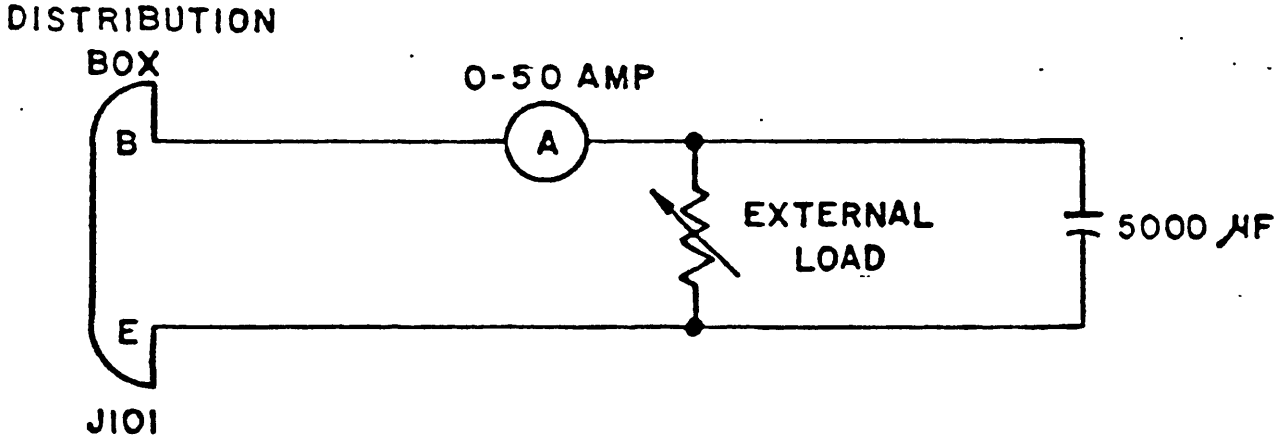


FIGURE 2

Battery Charging Current Limit Circuit

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3.19 Workmanship. All parts and assemblies shall be fabricated, and assembled in a thorough workmanship reamer. They shall be free of burrs,sharp edges, cracks, scratches, dents folds, wrinkles, buckles, dirt, grease, oil, rust, and other foreign matter. The cleaning method used shall not be injurious to any parts, nor shall the parts be contaminated by the cleaning agents. Exterior surface coatings shall be continuous; however, light scratches not exposing base material may be permitted. All required marking and-stamping shall be neat and clearly defined. Workmanship requirement. Requirement 9 of MIL-STD-454 shall apply.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of-all inspection requirements (examinations and tests) as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance-of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to ensure supplies and services conform to prescribed requirements.

4.1.1 Responsibility for compliance. All items shall meet all requirements of sections 3 and 5. The inspections set forth in this specification shall become a part of the contractor's overall inspection system or quality program. The absence of any inspection requirements in the specification shall not relieve the contractor of the responsibility of ensuring that all products or supplies submitted to the Government for acceptance comply with all requirements of the contract. Sampling inspection, as part of manufacturing operations, is an acceptable practice to ascertain conformance to requirements, however, this does not authorize submission of known defective material, either indicated *or* actual, not does it commit the Government to accept defective material.

4.1.2. General provisions. The component and subassembly inspection requirements of MIL-F-13926 form a part of the quality assurance provisions of this specification. Reference shall be made to MIL-STD-109 to define quality assurance terms used herein.

4.2 Classification of inspections. The inspection requirements specified herein are classified as follows:

a. First article inspection (see 4.2.1).
b. Quality conformance inspection (see 4.2.2).

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4.2.1 First article inspection. When required (see 3.1), First Article inspection shall be applied to the preproduction model or initial production item noted in 3.1. First article Inspection shall consist of the examination in 4.3, all tests under 4.4, test methods in 4.6.1, and verification of contractor inspection and test procedures in 4.6.2. Failure of the first article to pass the examination or any of the tests shall be cause for rejection.

4.2.2 Quality conformance inspection. Quality conformance inspection shall be applied to each production unit offered for acceptance under the contract. Quality conformance inspection shall consist of (a) through (c), as follows, and failure of a production unit to pass an examination or test shall be cause for rejection of the unit:

a. Product examination (see 4.3).
b. Functional tests (see 4.4.1 through 4.4.9).
c. preparation for delivery Inspection (see 4.5).

4.3 Product examination. Each production hydraulic power supply and distribution box shall be examined to determine compliance with the requirements of the contract. Failure to pass any examination shall be cause for rejection. The power supply shall be examined in accordance with the contractor's inspection procedures (see 4.6.2) to include, as a minimum, inspection of the following product characteristics.

4.3.1 Product characteristics. Material, construction and design not as specified (see 3.2). The hydraulic power supply (Drawing F12007000) and the distribution box (Drawing F12007200) shall be examined to verify configuration, identity and completeness of components, proper assembly, protective finish, hardware and marking as specified on the drawings.

Pump, not as specified (see 3.4.1).
Filters not as specified (see 3.4.2 through 3.4.2.2).
High pressure relief valve not as specified (see 3.4.3).
Reservoir not as specified (see 3.4.4). .
Oil cooler not as specified (see 3.4.6).
Accumulator not as specified (see 3.4.7).
Flow meter not as specified (see 3.4.8).
Thermal switch not as specified (see 3.4.9).
Electric system not as specified (see 3.5 through 3.5.3).
Battery not as specified (see 3.5.4).
Distribution box not as specified (see 3.6).
Internal accessibility not as specified (see 3.7).
Nameplate not as specified (see 3.9).
Electrical components not as specified (see 3.10).
Fungus control not as specified (see 3.11).
Wiring not as specified (see 3.12).
Soldering not as specified (see 3.13).
Identification of product not as specified (see 3.15).
Technical manuals not as specified (see 3.16).
Workmanship not as specified (see 3.18).

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4.4 Tests.

4.4.1 Hydraulic pressure and flow test.

a. Connect power supply AC power cable to 220 VAC 3 phase 50 to 60 Hertz supply and connect a hydraulic pressure gage, flow meter, and a variable external test load between the supply and return port. Close the outlet shutoff valve and open the bypass valve before energizing the power supply.
b. With master switch in the off position, verify that the start switch (S2) will not energize the hydraulic system. Place master switch in the on position and depress start button. Verify that pump motor (B1) and heat exchanger fan (B2) are energized. Red indicator lamp (DS2) should not be illuminated.
c. Adjust external variable hydraulic *test* load for 0 gpm flow rate, open outlet shutoff valve and close bypass valve. Adjust hydraulic pump compensator control to verify that pressure is continuously variable from 500 to 1500 psi. Verify that power supply hydraulic pressure gage reads within ± 50 psi of the test pressure gage at 500, 1000 and 1500 psi. Retain setting of 1500 psi.
d. Set the variable Hydraulic external test load for a flow of 4 gpm or greater. Adjust hydraulic pump volume control to verify flow is continuously available from 1 to 3.5 gpm. Verify that the power supply hydraulic flow meter reads within 2 percent of test flow meter at 1, 2, and 3.5 gpm. Retain setting of 3.5 gpm and lock pump volume control.
e. Shut outlet shutoff valve and observe 0 gpm flow through external flow meter and minimum indication on power supply flow meter. Open bypass valve and observe 3.5 gpm flow on power supply flow meter. Stop power supply.
f. Inspect power supply for external hydraulic leakage. External leakage of any component shall not exceed that specified in 3.4.5.

4.4.2 Pressure relief valve test. The test shall be performed using either Test Method A or Test Method B below. The test method shall be optional with the contractor except Method B shall be used only if the power supply is equipped with a pump which is adjustable to sufficient pressure. After testing, inspect power supply for external hydraulic leakage. External leakage of any component shall not exceed that specified in 3.4.5.

a. Test Method A: With power supply cable disconnected, connect an external hydraulic pressure source, test pressure gage, and test flow meter to the power supply port. Open outlet shutoff valve and close bypass valve. Increase pressure supplied from external hydraulic source

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until flow through test flow meter indicates actuation of high pressure relief valve. observe pressure indicated on test pressure gage to verify that the relief valve complies with 3.4.3.

b. Test Method B: With power supply AC supply cable connected, and with outlet valve closed and by-pass valve opened, operate power supply. Close by-pass valve and increase pump pressure compensator until pressure gage indicates actuation of the relief valve. Observe pressure indicated on power supply gage to verify that the relief valve complies with 3.4.3.

4.4.3 High pressure filter switch test. The test shall be performed using either Test Method A or Test Method B below. The test method shall be optional with the contractor.

a. Test Method A. With the power supply operating, simulate a pressure differential across the high pressure filter until the differential pressure switch is actuated and the red indicator light DS2 is illuminated. Measure the pressure differential to verify that the differential pressure switch complies with 3.4.2.

b. Test Method B: Each high pressure differential pressure switch shall be bench tested by introducing a differential pressure to verify conformance with 3 .4.2, prior to installation.

4.4.4 Thermal switch test. With the power supply operating at 1500 psi and 3.5 gpm, slowly raise the fluid temperature until thermal switch (S3) is actuated. Measure the fluid temperature and verify that the thermal switch complies with 3.4.9 and shuts down the hydraulic pump and heat exchanger fan.

4.4.5 DC voltage regulation tests.

a. With the power supply connectors P105 and P104 connected to the distribution box receptacles J105 and J104, connect a test ammeter and variable load between pin K and pins H and Y J102. Adjust variable load for maximum resistance.

b. Apply 208 Volts AC 3 phase 50-60 Hertz to power cart AC supply cable. With the master switch in the off position and the selector switch (S5) in the DC power supply position, verify O VDC reading on voltmeter (V) with selector switch (SI) in the 28 VDC buss position.

c. Turn on master switch and adjust the DC voltage adjust for 30 VDC as indicated on a test voltmeter attached between TJBP1 and TJBP4 on the distribution box. Verify that voltmeter (V) is accurate to within 2 percent of full scale as compared with test voltmeter.

d. Turn selector switch (S5) to off, and verify DC meter reading returns to 0 VDC. Place selector switch (S5) in DC power supply position. Allow voltage regulator of power supply to warm up for 10 minutes. Decrease

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variable load resistance until test ammeter indicates 40 amps and verify that voltage regulation is ± 1 percent or better at the power supply TBI-3 and TBI-4, over the load range. While load is being varied, verify that ammeter (A) is accurate to within 2 percent of full scale as compared with test ammeter when selector switch (S3) is in the 50 amp position and the press to read switch (SW2) is depressed. Connect an oscilloscope or peak to peak meter between TJBP2 and TJBP5 and verify peak to peak ripple does not exceed 200 millivolts at full load.

e. Adjust variable load for maximum resistance. Adjust DC voltage for 26 volts on distribution box voltmeter and verify that voltmeter (V) is within 2 percent of full scale, as compared with test voltmeter. Decrease load resistance until test ammeter Indicates 40 amps and verify that voltage regulation is within ± 1 percent at the power supply (TBI-3 and TBI-4). Verify that peak to peak ripple does not exceed 200 millivolts. Adjust variable load for maximum resistance.

f. Apply 230 VAC 3 phase 50 - 60 hertz to power cart AC supply cable and repeat paragraphs c through e. above.

g. Adjust DC voltage output for 30 VDC and adjust variable load for 65 amp output until 50 amp circuit breaker trips, then adjust to maximum resistance. Verify that DC breaker will reset.

h. Utilizing test voltmeter verify that 30 VDC is present at the following points:

(1) TJBP2 to TJBP5
(2) TJBP3 to TJBP6
(3) Pin A to Pin B on J106 and J107
(4) Receptacle A to B on J103 (both positions)
(5) Pin P, O, R, J to Pins H and Y of J102

i. Verify continuity from TJ1 to Pin E of J102 and TJ2 to Pin F of J102.

4.4.6 DC supply current limit test.

a. With power supply AC power cable disconnected, connect an external 24 volt source, external load, and ammeter to TB2 as shown in Figure 1.

b. Connect power supply AC power cable to 220 VAC 3 phase 50 to 60 Hertz supply. Connect power supply connectors P105 and P104 to distribution box receptacles J105 and J104. Energize power supply and adjust output to 28 VDC, distribution box voltmeter can be used.

c. Starting with external load resistance set at maximum, increase current through resistor (R10) until start of current limit and continue until voltage output is 1 volt maximum. Verify that observed ammeter readings are in compliance with 3.17.4.b.(3).(b).

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4.4.7 DC battery buss test.

a. Disconnect power supply from AC source and install a charged DC battery as specified in 3.5.4 and 4.6.1.4. This battery shall be one used for test purposes only and not one of those to be supplied with each production unit. (See 3.5.4). With power supply DC output cables attached to J104 and J105 on the distribution box, attach a variable load, set for maximum resistance, and a test ammeter across pins B to E of J101. Place selector switch (SW1) in the Battery Buss position and verify that battery voltage is indicated on voltmeter (V).
b. With selector switch (SW3) in the 150 amp position and SW2 depressed, set load for 60 amps and 90 amps and verify that ammeter (A) is within 2 percent of full scale as compared with a test ammeter.
c. With selector switch (SW3) in the 500 amp position and SW2 depressed, set load for SO amps and 100 amps and verify that ammeter (A) is within 2 percent of full scale as compared to a test ammeter.
d. Upon completion of testing, remove battery from power supply.

4.4.8 Battery charge voltage test.

a. Connect power supply AC power cable to 220 VAC 3 phase 50 to 60 Hertz supply. Connect power supply connector P104 to the distribution box. Connect a test ammeter, external load, and 5,000 MFD Capacitor between pins B and E of connector J101 as shown In Figure 2. Place the voltmeter selector switch on the distribution box to the Battery Buss Position.
b. Energize the power supply with the control panel selector switch in the battery charge position and the DC voltage adjust in the maximum clockwise position. Increase the load. The current limit shall occur between 7 to 13 amps. Verify that the power supply ammeter is within 2 percent of full scale as compared with a test ammeter.

4.4.10 Tie down test. The tie down connections on the portable hydraulic power supply shall be subjected to the loads specified in 3.14.1. There shall be no evidence of weld failure or permanent deformation of the connections. Scientific computations and analysis may be submitted in lieu of actual test to verify compliance with specified requirements. Such analytical proof may, at the option of the Government, be subject to verification by actual test.

4.4.11 Electromagnetic interference test. The hydraulic power supply shall be tested in accordance with MIL-STD-461 for IIB equipment, Table II, to verify compliance to 3.8.

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4.4.12 Environmental tests. The power supply and distribution box shall be subjected to the following environmental tests. All data shall be recorded on continous recording type charts capable of being read with a resolution of 1 deg. F. Prior to conducting the high and low temperature storage test and the non operating altitude test (15,000 ft.), the accumulator shall be repressurized. Prior to conducting the low storage test, remove the battery.

4.4.12.1 High temperature operating test. The power supply and distribution box shall be stabilized in an ambient air temperature of plus 125°F. There shall be no evidence of damage to any component or leakage exceeding that specified in 3.4.5. While at that temperature, the power supply shall be tested as specified in 4.4.1 through 4.4.9.

4.4.12.2 Low temperature operating test. The power supply and distribution box shall be stabilized in an ambient air temperature of 0°F. There shall be no evidence of damage to any component or leakage exceeding that specified in 3.4.5. While at that temperature, the power supply shall be tested in accordance with 4.4.1 through 4.4.9.

4.4.12.3 High temperature storage test. After depressurizing the accumulator, the hydraulic power supply and distribution box shall be placed in an ambient air temperature of plus 160 deg. F for a period of 72 hours. At the end of that time, the power supply shall be removed and inspected. There shall be no evidence of damage to any component or leakage exceeding that specified in 3.4.5. Following stabilization at plus 80 F deg. +10 deg. F, the power supply shall be tested in accordance with 4.4.1 through 4.4.9.

4.4.12.4 Low temperature storage test. After repressurizing the accumulator, the hydraulic power supply and distribution box shall be placed in an ambient air temperature of minus 65 deg. F for a period of 72 hours. At the end of that time, the power supply shall be removed and inspected. There shall be no evidence of damage to any component or leakage exceeding that specified in 3.4.5. Following stabilization at plus 80 deg. F, $\pm$ 10 deg. F, the power supply shall be tested in accordance with 4.4.1 through 4.4.9.

4.4.12.5 Humidity test. The hydraulic power supply and distribution box shall be subjected to 24 hours exposure at a relative humidity between 95 and 100 percent and at an air temperature of plus 125 deg. F. The power supply and distribution box control panels shall be exposed (see 3.3c) and connected for test as specified in 4.4.5. While at this condition, the power supply shall be tested in accordance with 4.4.1 through 4.4.9.

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4.4.12.6 Altitude test. The hydraulic power supply and distribution box shall be placed in a temperature-altitude chamber equivalent to 6,000 feet (above sea level) at a temperature of 0 deg. F. There shall be no evidence of damage to any component or leakage exceeding that specified in 3.4.5. While at this condition, the hydraulic power supplly and distribution box shall be tested in accordance with 4.4.1 through 4.4.9. The temperature in the chamber shall be raised to plus 125 deg. F and the tests in 4.4.1 through 4.4.9 repeated. Upon satisfactory completion of these tests, repressurize the accumulator and raise the pressure in the chamber equivalent to 15,000 feet at an air temperature of 0 deg. F. Maintain this condition for a period of two hours. There shall be no evidence of damage to any component or leakage exceeding that specified in 3.4.5. The temperature of the chamber shall then be raised to plus 125 deg. and maintained for a period of 2 hours. There shall be no evidence of damage to any component or leakage exceeding that specified in 3.4.5. The chamber shall be returned to ambient conditions, and the hydraulic poweer supply and distribution box shall be tested in accordance with 4.4.1 through 4.4.9.

4.4.12.7 Salt fog test. The hydraulic power supply and distribution box shall be subjected to the salt fog test specified in MIL-STD-810, Method 509. In reference to the pretest performance record required in paragraph 3.2.1 of MIL-STD-810, satisfactory completion of the tests specified herein in 4.4.1 through 4.4.9 during or upon completion of any of the preceding environmental tests (4.4.12), shall serve as the pretest performance record. The hydraulic power supply and distribution box control panels shall be exposed (see 3.3d) and connected for test as specified in 4.4.5 Following this test, the hydraulic power supply and distribution box shall be tested in accordance with 4.4.1 through 4.4.9.

4.4.13 Fluid contaminant test. Following the completion of all other hydraulic power supply tests specified herein, perform the following tests:

a. Verify that the power supply pump is adjusted for 1500 psi and 3.5 gpm output in compliance with 3.17.3, a. and b.
b. The hydraulic power supply shall be tested to verify compliance with the maximum contaminant limits specified in 3.4 If the specified limits are not met, the power supply shall be cleaned and flushed in accordance with Drawing 11011289, a new filter installed, refilled with fluid in accordance with MIL-H-50606, and the test shall be repeated. Contaminant limits still above those specified shall constitute failure of this test.

4.5 preparation for delivery inspection. Preservation, packaging, packing and marking shall be inspected to verify compliance with Section 5.

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4.6 Methods of inspection.

4.6.1 Test methods.

4.6.1.1 Power Supply. The power supply shall be tested in accordance with the following methods and the contractor's test procedures in 4.6.2. Where referenced in tests specified herein, the term "hydraulic power supply" defines Power Supply Hydraulic/Electric, Portable, Drawing 12007000. The term "distribution box" defines Hydraulic/Electric Distribution Box and Accessories, Drawing 12007200.

4.6.1.2 Controls and operation. Prior to beginning tests, all control adjustments and operations shall have been accomplished on the hydraulic power supply and distribution box as specified in 3.17.

4.6.1.3 Cleaning. Prior to, and at the conclusion of tests specified in 4.4, the hydraulic power supply shall be cleaned and flushed in accordance with Drawing 11011289. (see 3.17).

4.6.1.4 Preparation for use. Prior to testing, the hydraulic power supply and distribution box shall be prepared for use as specified in TM 9-4933-211-14, Section II, paragraphs 2.3 through 2.6 and 3.17.1 herein. A battery (see 3.5.4) shall be installed and charged in accordance with the instructions in paragraph 2.6 of the TM and the instructions contained on the battery.

4.6.1.5 Instruments. Instruments used in conjunction with testing specified herein shall have an accuracy traceable to the National Bureau of Standards and shall be calibrated in accordance with MIL-STD-45662.

4.6.1.6 Temperature. Temperature shall be measured by dial thermometers or thermocouples accurate within one percent of the measured value. Thermocouples shall be Insulated from contact with other metals.

4.6.1.7 Pressure. Pressures shall be measured by calibrated Bourdon tube gages and shall be accurate within one percent of the measured valve.

4.6.1.8 Current and voltage. Current and voltage shall be measured with instruments accurate within one half of one percent of the measured values.

4.6.1.9 Liquid flow rates. Liquid flow rates shall be measured with a calibrated flowmeter having an accuracy of one percent of the measured value. Instantaneous flowmeter readings shall not be used.

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4.6.2 Contractor prepared inspection and test procedures.

4.6.2.1 Descriptive list. The contractor shall prepare a descriptive list of inspection and test equipment required for complete checkout of the hydraulic power supply and distribution box. Primary characteristics of the test equipment shall be given, such as input, output, range(s), accuracy, etc.

4.6.2.2 Inspection checklist. The contractor shall prepare an inspection check list of components, subassemblies and assemblies to be inspected, cross referenced to applicable requirement drawings. For each item listed, product characteristics to be inspected shall be given. As a minimum, examples of such characteristics to be inspected are:

a. Completeness of components.
b. Proper assembly (type and quantity of components).
c. Application and type of protective finish.
d. Identity and installation of all controls, marking and application of control plates schematics, etc.
e. Fastening devices (quantity, type, secureness).
f. Proper wiring, wiring connections, solder joints, lead dress and routing.
g. Proper make-up and assembled length of leads and harnesses.
h. Location of mounting brackets and holes in sub-assemblies.
i. Application of fungus treatment.
j. Physical characteristics of meters and controls (mounting dimensions, ranges, dial face, etc.).
k. Proper vendor part number identification for source and specification control drawing items.
l. Proper modifications shown on altered item drawings.
m. Workmanship (reference, Requirement 9 of MIL-STD-454).
n. Soldering (reference, Requirement 5 of MIL-STD-454).
o. Individual component dimensions and characteristics that affect function of the power supply and interchangeability of components.
p. Proper make-up and routing of hydraulic system tube and hose assemblies, lines and connections, proper assembly of hydraulic components and controls.

4.6.2.3 Test procedures. The contractor shall prepare detailed test procedures to be used in conducting all tests specified herein. Procedures shall include test methods, test equipment, test setups with schematics, etc. Individual component or bench testing shall be performed on components to verify performance requirements contained in drawing notes-prior to final assembly. Component performance requirements such as electrical characteristics, proof pressure, temperature, etc. shall be substantiated by vendor certification or test data, as applicable. Components (e.g. switches, controls, valves, etc.) requiring operation over specified ranges shall be tested to verify operation at the specified min-max conditions. The contractor shall

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prepare a list of these items indicating detailed tests to be performed and the test equipment required. Continuity checks (pin to pin, pin to terminal, terminal to terminal, etc.) shall be performed on all electrical cables.

4.6.2.4 Acceptability. The acceptability of the above contractor prepared inspection and test procedures shall be verified during First Article tests. They shall be included as part of the First Article test report, and shall be subject to Government approval under the First Article provisions of the contract. Upon approval of these procedures, the Government reserves the unrestricted right to the procedures for the purposes of conducting or specifying those inspections and tests during rebuild, future procurement actions etc.

4.6.2.5 Test sequence. Tests need not be run in the order listed except for the fluid contaminant test in 4.4.13. This test shall be the last to be conducted as part of the specified First Article tests.

5. PACKAGING

5.1 Pilot pack. The pilot pack shall consist of a complete power supply unit (see 1.1) packaged in accordance with the requirements specified herein and for the level of protection specified in the contract (see 6.2), packed level C, and forwarded in accordance with 3.1.

5.2 Preservation and packaging. Preservation and packaging shall be level A, C, or commercial as specified (see 6.2).

5.2.1 Level A.

5.2.1.1 Cleaning and drying. The power supply unit and all accessories shall be cleaned and dried in accordance with MIL-P-116. Cleaning shall be in accordance with process C-1 and drying shall be in accordance with any applicable procedure specified therein providing cleanliness established on applicable drawings for the hydraulic system is maintained.

5.2.1.2 Preservative application. Application of all preservatives shall be in accordance with the applicable drawing requirements maintaining requirements for the hydraulic system.

5.2.1.3 Unit packaging.

5.2.1.3.1 Power supply (less battery) The power supply (less battery) shall be packaged in accordance with Method I of MIL-P-116.

5.2.1.3.2 Distribution box. The distribution box shall be packaged in accordance with Method II of MIL-P-116.

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5.2.1.3.3 Battery. The battery shall be packaged in accordance with MIL-P-6063.

5.2.2 Level C.

5.2.2.1 Cleaning and drying. Cleaning drying shall be the same as specified for level A.

5.2.2.2 Preservative application. Preservative application shall be the same as specified for level A.

5.2.2.3 Unit Packaging.

5.2.2.3.1 Power supply (less battery) The power supply (less battery) shall be packaged the same as specified for level A.

5.2.2.3.2 Distribution box. The distribution box shall be packaged in accordance with Method III of MIL-P-116.

5.2.2.3.3 Battery. The battery shall be packaged in accordance with MIL-P-6063.

5.2.3 Commercial. The power supply shall be packaged in accordance with the applicable requirements of ASTM-D3951.

5.2.4 Repair parts. Repair parts shall be preserved, packaged, packed, and marked in accordance with the Packaging Data Sheet, or other requirements, and for the level of protection specified in the contract (see 6.2).

5.3 Packing. Packing shall be level A, C or commercial.

5.3.1 Level A. The complete unit shall be packed in a full sheathed crate conforming to MIL-C-104. The arrangement of contents, cushioning, blocking and bracing, and anchoring shall be in accordance with MIL-STD-1186.

5.3.2 Level C. The complete unit shall be packed in accordance with the applicable requirements of MIL-P-14232.

5.3.3 Commercial. The complete unit shall be packed in accordance with the applicable requirements of ASTM-D3951.

5.4 Marking.

5.4.1 Marking. In addition to any special markings specified herein, interior packages and exterior containers shall be marked in accordance with MIL-STD-129.

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5.4.1.1.1 Special marking for the hydraulic system. The following marking shall be applied to the hydraulic system in accordance with MIL-STD-129: "Do not open until ready for use. Cleaned for hydraulic service. If interior is contaminated reprocess IAW drawing requirements."

6. NOTES

(This section contains information of a general or explanatory nature that may be helpful, but is not mandatory.)

6.1 Intended use. The power supply is intended to provide hydraulic fluid supply and regulated DC power supply for ground distribution to aircraft weapon systems.

6.2 Acquisition requirements. Acquisition documents must specify the following:

a. Title, number and date of this specification.
b. First article, when required (see 3.1 and 4.2.1).
c. Shipping instructions for the first article and-pilot pack (see 3.1, 4.2.1 and 5.1).
d. Block of serial numbers (see 3.15).
e. Selection of applicable levels of preservation, packaging and packing (see 5.2 and 5.3).
f. Packaging instruction for repair parts, when applicable (See 5.2.4).
g. Issue of DODISS to be cited In the solicitation, and if required, the specific issue of individual documents referenced (see 2.1.1).

6.2.2 Contract data requirements. The requirements for records of examination and all tests will be listed directly on a DD Form 1423 incorporated into the contract.

6.2.3 Management control system requirements. When warranted, the contract should specify the application of MIL-Q-9858 or MIL-I-45208 on the Management Control Systems Summary List, DD Form 1660.

6.2.4 Management control system's summary list. When warranted, the contract should specify the application of MIL-I-45607 and MIL-STD-45662 on the Management Control Systems Summary List, DD Form 1660.

6.3 National stock number. The power supply has been assigned National Stock Number NSN 4933-00-933-4742.

6.4 Submission of inspection equipment designs for approval. Submit equipment designs as required to Commander, ARDEC ATTN: AMSMC-QAF-I (D), Picatinny Arsenal, NJ 07806-5000. This address will be specified on the Contract Data Requirements Lists, DD Form 1423 in the contract. Unless otherwise specified, data item DI-R-1714 will apply.

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6.5 Submission of results of contractor conducted examinations and tests. Unless otherwise specified by the contracting officer, the contractor should forward requested records of examinations or tests to Commander, ARDEC ATTN: AMSMC-QAF-A (D), Picatinny Arsenal, NJ 07806-5000.

6.6 Process deviation. A process deviation is defined as a change in the approved basic method of manufacture, or an operational change which alters the metallurgical or physical properties of the item.

6.7 Drawings. Drawings listed in Section 2 of this specification under the heading US Army Armament Research, Development and Engineering Center (ARDEC) may also include drawings prepared by, and identified as, Edgewood Arsenal, Frankford Arsenal, Rock Island Arsenal, Picatinny Arsenal, US Army Armament Research and Development Command (ARRADCOM) or US Army Armament Research and Development Center (ARDC) drawings, Technical data originally prepared by these activities is now under the cognizance of ARDEC.

6.8 Subject term (Key word) listing.

Dry Nitrogen

6.9 Changes from previous issue. Marginal notations are not used in this revision to identify changes with respect to the previous Issue due to the extensiveness of the changes.

Custodian:
Army-AR

Preparing activity:
Army-AR

(Project 4933-A033)

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

INSTRUCTIONS

1. The preparing activity must complete blocks 1, 2, 3, and 8. In block 1, both the document number and revision letter should be given.
2. The submitter of this form must complete blocks 4, 5, 6, and 7.
3. The preparing activity must provide a reply within 30 days from receipt of the form.

NOTE: This form may not be used to request copies of documents, nor to request waivers, or clarification of requirements on current contracts. Comments submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or to amend contractual requirements.

I RECOMMEND A CHANGE	1. DOCUMENT NUMBER	2. DOCUMENT DATE (YYMMDD)
	MIL-P-45969B (AR)	24 August 1990

3. DOCUMENT TITLE
POWER SUPPLY, HYDRAULIC AND ELECTRIC, PORTABLE

4. NATURE OF CHANGE (*Identify paragraph number and include proposed rewrite, if possible. Attach extra sheets if needed.*)

5. REASON FOR RECOMMENDATION

6. SUBMITTER

(1) Commercial
(2) AUTOVON
(If applicable)

8. PREPARING ACTIVITY

a. NAME
US ARMY ARDEC
STANDARDIZATION OFFICE

b. TELEPHONE (*Include Area Code*)
(1) Commercial (201) 724-6662
(2) AUTOVON 880-6662

c. ADDRESS (*Include Zip Code*)
ATTN: SMCAR-BAC-S
PICATINNY ARSENAL, NJ 07806-5000

IF YOU DO NOT RECEIVE A REPLY WITHIN 45 DAYS, CONTACT:
Defense Quality and Standardization Office
5203 Leesburg Pike, Suite 1403, Falls Church, VA 22041-3466
Telephone (703) 756-2340 AUTOVON 289-2340

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